

① Find $\frac{dy}{dx}$ if $(\cos x)^y = (\cos y)^x$.

② Find the particular solution of the differential equation $\frac{dy}{dx} = -\left(\frac{x+y\cos x}{1+\sin x}\right)$ given that $y=1$, when $x=0$.

③ If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$ and $A^3 - 6A^2 + 7A + KI_3 = 0$, find k

④ Evaluate $\int_{-1}^2 |x^3 - x| dx$

⑤ Find the equations of tangent to the curve $y = x^3 + 2x - 4$, which are perpendicular to the line $x + 4y + 3 = 0$.

⑥ Find the image of the point $(2, -1, 5)$ in the line $\frac{x-1}{10} = \frac{y+2}{-4} = \frac{z+8}{-11}$. Also, find the equation

of the line joining the given point and its image. Find the length of that line-segment also.

⑦ Prove that $y = \frac{4\sin\theta}{2+\cos\theta} - \theta$ is an increasing function in $[0, \pi/2]$

⑧ Evaluate $\int \frac{x^2+1}{x^4+1} dx$

⑨ Evaluate $\int \frac{x^2}{(x\sin x + \cos x)^2} dx$

⑩ Evaluate $\int \frac{\sin x - \cos x}{\sqrt{\sin 2x}} dx$